

Warm-Start Diffusion for Low-Resolution Probabilistic ERA5 Weather Forecasting

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Abstract

Probabilistic weather forecasting needs both good forecast scores and physically reasonable samples. In this project, I test a simple hybrid for low-resolution ERA5-MINI: first a U-Cast-style Monte Carlo dropout U-Net makes a forecast, then an EDM diffusion model is warm-started from that forecast and denoises only at lower noise levels. Averaged across evaluated timestamps, U-Cast gets CRPS/RMSE of 86.798/202.97, cold-start diffusion gets 89.26/211.03, warm-start diffusion improves this to 85.23/201.94, and fine-tuned warm-start diffusion reaches 84.98/201.59. The result supports the main idea: CRPS training is good at forecast placement, while diffusion is useful as a more physical low-noise refiner.

1 Introduction

Weather forecasting is a hard probabilistic generation problem because the outputs need to score well and still look like actual weather. In this project, I use ERA5-MINI, a coarse 64×32 global ERA5/WeatherBench2-style grid, so that I can test the idea without needing full operational-scale training. The model predicts 12-hour transitions from two previous states and is evaluated over 15 autoregressive steps.

The issue I am trying to get at is the tradeoff between direct probabilistic training and diffusion generation. CRPS-trained systems such as FGN, AIFS-CRPS, and U-Cast optimize forecast skill directly and are efficient at inference [1, 12, 3]. Diffusion models learn a denoising generative process and often produce better spatial structure, but cold-start sampling begins far from the data manifold and requires many denoising evaluations [8, 13, 2]. So I test whether diffusion is more useful as a local refiner: start from a strong CRPS forecast, add controlled noise, and denoise only the lower-noise tail of the EDM schedule.

The main pieces of the project are: (1) a U-Cast-style ERA5-MINI baseline trained with weighted MAE then weighted CRPS and Monte Carlo dropout; (2) a conditional EDM baseline with the same Dhariwal-style U-Net backbone; (3) a warm-start sampler and fine-tuning setup using the CRPS model as the proposal;

and (4) evaluation using aggregate CRPS/RMSE, lead-time skill curves, warm-start-step ablation, and spectra. The observed ordering is fine-tuned warm-start EDM > warm-start EDM > U-Cast-style CRPS $\gg$ cold-start EDM. The effects are slight but consistent, and support the hypothesis that CRPS gives strong forecast placement while diffusion supplies a more physical local prior.

2 Related Work

GraphCast showed that neural models can make skillful medium-range forecasts from ERA5 [11], while ERA5 and WeatherBench2 provide the reanalysis and benchmark context for this project [7, 14]. I am not trying to reproduce a full GraphCast-style system here. Instead, I use a small ERA5-MINI setting whose variables follow the parsimonious modeling spirit of DLWP-HPX [9]. DLWP-HPX uses Z_{1000} , Z_{500} , $Z_{300} - Z_{700}$ thickness, T_{2m} , T_{850} , $TCWV$, and Z_{250} ; here I keep Z_{300} and Z_{700} separate, giving eight predicted channels.

Proper scoring rules such as CRPS are standard for probabilistic forecasting [6]. U-Cast is the closest baseline: a simple U-Net is pretrained deterministically, fine-tuned with CRPS, and sampled using Monte Carlo dropout [3, 5]. Diffusion models instead reverse a noise process [8]; I use an EDM formulation [10] and a Dhariwal/Nichol U-Net family [4, 15]. Weather diffusion systems such as GenCast and cBottle motivate diffusion for sample quality [13, 2], while fast sampling and warm-start work motivate avoiding unnecessary high-noise denoising when a strong conditional proposal is available [16, 17]. The gap I focus on is therefore specific: direct CRPS training gives strong scalar skill, but diffusion may add a spectral/sample-quality prior if it is used only where it is most reliable.

3 Method

Forecasting setup and architecture. Let $x_t \in \mathbb{R}^{C \times H \times W}$ be the normalized ERA5-MINI state, with $C = 8$ channels and $(H, W) = (64, 32)$:

$$\{T_{2m}, Z_{250}, Z_{300}, Z_{500}, Z_{700}, Z_{1000}, T_{850}, TCWV\}.$$

Inputs are (x_{t-12h}, x_t) plus land-sea mask, surface geopotential, and sine/cosine day/year progress. Losses use latitude weighting by $\cos(\phi)$. The runs train on 1979–2019 slices and evaluate in validation mode with up to 128 samples and 10 ensemble predictions per initial condition. All variants use the same Dhariwal-style convolutional U-Net: 256 base channels, channel multipliers [1, 2, 4], residual prediction, dropout 0.1, attention at lower resolutions, and circular longitude padding.

U-Cast-style baseline. The deterministic stage predicts a residual update and minimizes weighted MAE,

$$\mathcal{L}_{\text{WMAE}} = \mathbb{E} \sum_{c,i,j} w_{c,i,j} |f_{\theta}(x_{t-12h}, x_t)_{c,i,j} - x_{t+12h,c,i,j}|.$$

The probabilistic stage enables dropout and minimizes weighted ensemble CRPS:

$$\text{CRPS}(\{y^{(m)}\}, x) = \frac{1}{M} \sum_m \|y^{(m)} - x\|_1 - \frac{1}{2M^2} \sum_{m,n} \|y^{(m)} - y^{(n)}\|_1.$$

The main CRPS model uses two training ensemble members and 10 Monte Carlo dropout samples at evaluation. This same model also supplies the warm-start proposal for diffusion.

Cold-start EDM. The diffusion baseline follows EDM [10]. For context $c = (x_{t-12h}, x_t)$, sample σ and $\epsilon \sim \mathcal{N}(0, I)$, corrupt the target as $z = x + \sigma\epsilon$, and train D_{θ} with

$$\mathcal{L}_{\text{EDM}} = \mathbb{E}_{x,c,\sigma,\epsilon} \lambda(\sigma) \|D_{\theta}(x + \sigma\epsilon, c, \sigma) - x\|_2^2.$$

The baseline uses $\sigma_{\min} = 0.002$, $\sigma_{\max}^{\text{inf}} = 800$, $\rho = 7$, 20 sampler steps, $P_{\text{mean}} = -1.6$, $P_{\text{std}} = 1.6$, weighted MSE, batch size 128, and 10k training steps. This model has realistic spectra but weaker direct skill, likely because the same denoiser has to handle both high-noise generation and low-noise forecast correction.

Warm-start EDM. Warm-start diffusion initializes from a CRPS forecast rather than from high noise. If $\hat{x}_{t+h}^{\text{UC}} \sim q_{\psi}(x_{t+h} \mid x_{t-12h}, x_t)$, I start the sampler from

$$z_{\sigma_K} = \hat{x}_{t+h}^{\text{UC}} + \sigma_K \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

then run the low-noise tail $z_{\sigma_K} \rightarrow \dots \rightarrow z_{\sigma_0}$. The best inference setup uses U-Cast guidance, 6 warm-start steps out of a 20-step EDM schedule, warm-start dropout, and 10 predictions. I also fine-tune the diffusion model for this regime, using $P_{\text{mean}} = -3$, $P_{\text{std}} = 1$, $\sigma_{\max}^{\text{inf}} = 400$, and per-sigma loss logging.

Baselines and metrics. The comparison includes deterministic WMAE U-Net, U-Cast-style WCRPS U-Net, cold-start EDM, warm-start EDM, and fine-tuned warm-start EDM. I focus on normalized CRPS, RMSE, lead-time skill curves, the warm-start step ablation, and energy spectra.

4 Results and Analysis

Experimental details. All main experiments use 12-hour resolution, window size 2, 15-step rollout, 256 model channels, [1, 2, 4] channel multipliers, residual

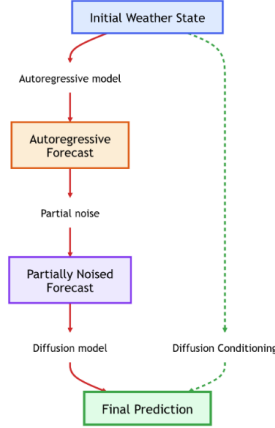


Figure 1: Warm-start diffusion pipeline. The autoregressive CRPS model proposes an initial forecast, partial noise is added, and the diffusion model refines the noised forecast while conditioning on the initial weather state.

Table 1: Aggregate forecast skill averaged across all evaluated timestamps. Lower is better for both metrics.

Model	CRPS ↓	RMSE ↓	Notes
U-Cast-style CRPS	86.798	202.97	Strong direct probabilistic baseline.
Cold-start EDM	89.26	211.03	Worse skill, but more physical spectra.
Warm-start EDM	85.23	201.94	Best non-fine-tuned hybrid.
Fine-tuned warm-start EDM	84.98	201.59	Best overall aggregate skill.

prediction, dropout 0.1, and latitude-weighted losses. The U-Cast-style baseline enables inference dropout. Cold-start diffusion uses 20 EDM steps with $\sigma_{\min} = 0.002$, $\sigma_{\max}^{\text{inf}} = 800$, and no warm start. Warm-start diffusion uses the same EDM model with U-Cast guidance, 6 warm-start steps, and 10 predictions. The CRPS model is initialized from the deterministic WMAE checkpoint.

Table 1 reports the aggregate metrics averaged across all evaluated timestamps. I leave out the deterministic MAE-only baseline because I do not have the corresponding aggregate CRPS/RMSE values for that model.

The completed runs show the ordering fine-tuned warm-start EDM > warm-start EDM > U-Cast-style CRPS $\gg$ cold-start EDM on the aggregate metrics. Relative to U-Cast, warm start improves CRPS by 1.57 and RMSE by 1.03, while fine-tuned warm start improves CRPS by 1.82 and RMSE by 1.38. Cold-start EDM is much worse on direct skill, which supports the view that the full high-to-low-noise denoising task is not well matched to this forecast metric. Warm start helps because the CRPS model supplies the forecast placement and diffusion only has to make a local correction near the data manifold.

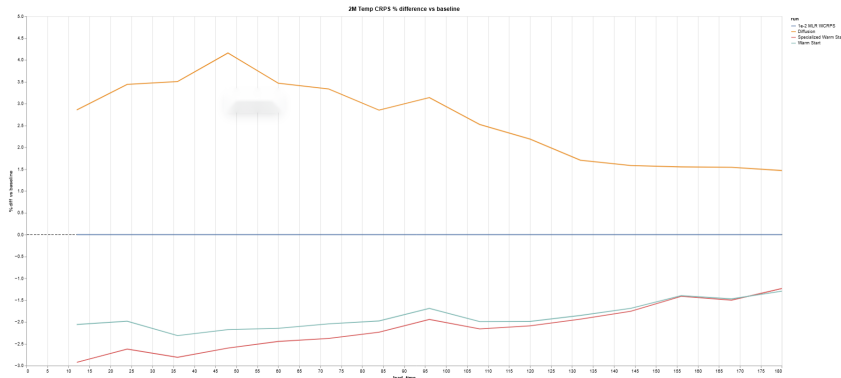


Figure 2: Lead-time CRPS percent difference from the U-Cast-style baseline for 2m temperature. Cold-start diffusion is worse than the baseline, while warm start and fine-tuned warm start improve consistently across lead times.

Qualitative and physical diagnostics. Rollout maps would be useful for representative variables such as T_{2m} , Z_{500} , T_{850} , and $TCWV$, but in this report the strongest evidence is from lead-time skill curves and spectral structure.

The energy spectrum diagnostic is central. For Z_{250} , U-Cast has excess high-wavenumber power relative to the ground truth, while diffusion and warm start track the ground-truth decay more closely. This explains the apparent tension in the scalar metrics: diffusion alone is not the best forecaster, but it learns a useful prior over spatial structure. Warm-start diffusion benefits from both sides: U-Cast supplies large-scale forecast placement, and low-noise EDM refinement nudges samples toward a more realistic atmospheric manifold.

Why warm start helps. I think the result is best understood as a division of labor. The CRPS model solves forecast placement; diffusion supplies a sample-manifold prior. Cold-start diffusion asks one model to do both from high noise. Warm start instead asks EDM to repair an already plausible forecast, and fine-tuning further specializes the denoiser to the low-noise regime actually used at inference. Figure 4 shows that 6 warm-start denoising steps gives the best CRPS/RMSE tradeoff; using too many steps moves the sampler closer to the weaker cold-start regime.

5 Limitations

There are a few important limitations. ERA5-MINI is deliberately low resolution, so this should not be read as evidence about operational-resolution forecasting. The comparison is also not compute-normalized: diffusion uses multiple U-Net evaluations per sample, while U-Cast is much cheaper. Finally, the current analysis is strongest for aggregate skill and spectra; future work should test higher resolution, more variables, and stricter test-set evaluation.

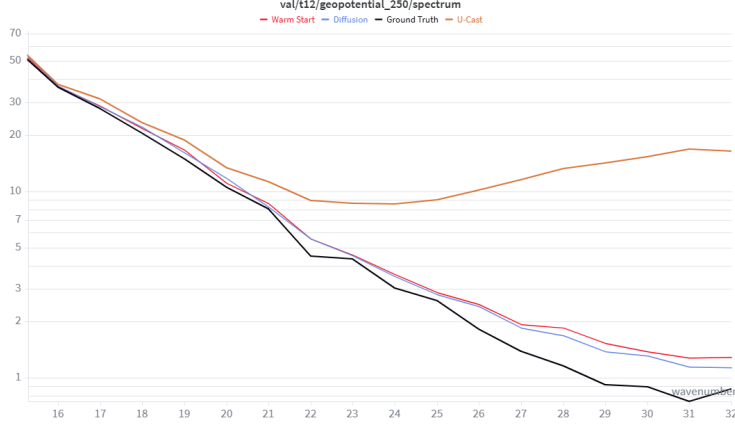


Figure 3: Energy spectrum for Z_{250} . U-Cast over-amplifies high wavenumbers, whereas diffusion and warm start are closer to the ground-truth spectral slope.

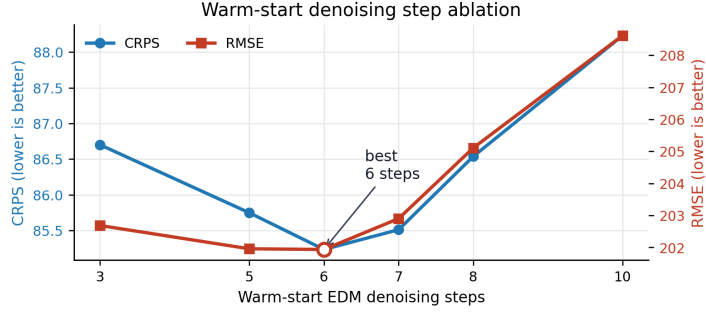


Figure 4: Warm-start step ablation. Six denoising steps gives the lowest CRPS and RMSE; both shorter and longer warm-start trajectories degrade the aggregate metrics.

6 Conclusion

Overall, this project suggests that diffusion is more useful for this setting when it starts from a strong direct forecast rather than from pure noise. U-Cast gives strong direct skill, cold-start EDM gives a more physical prior but weaker metrics, and warm-start EDM combines the two to give the best observed performance. The broader lesson is that forecast skill and physical sample quality are related but not identical. Warm-start diffusion is a practical way to combine CRPS-trained forecasting with diffusion-based refinement, and fine-tuning the denoiser for the warm-start regime appears most promising.

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